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Authenticated Gateway-Assisted DNSSEC for IoT Clients

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Problem Description

Clear outline of the problem being addressed

- **Memory & Computational Bottleneck:** Microcontrollers (e.g., ESP32) have ∼520 KB SRAM, making parsing multi-kilobyte RSA/ECDSA signature chains (`RRSIG`, `DNSKEY`) prohibitive.
- **Packet Fragmentation:** DNSSEC payloads routinely exceed standard UDP MTU sizes, causing packet drops over low-power wireless links.
- **The Security Dilemma:** Disabling DNSSEC exposes IoT devices to DNS spoofing and cache poisoning; enabling local DNSSEC causes Out-of-Memory (OOM) crashes or drains battery reserves.
- **Unauthenticated Last-Hop:** Non-validating IoT stub resolvers blindly trust upstream responses over unencrypted channels without authenticating the local gateway.

Background

Necessary context to understand the problem and optimization

- **DNSSEC Chain of Trust:** Authoritative DNS servers digitally sign records using Key Signing Keys (KSK) and Zone Signing Keys (ZSK). Resolvers recursively verify signatures up to root trust anchors.
- **IoT Operational Disparity:** Smart cities and industrial automation require secure backend domain lookup, but edge nodes operate on tight milliwatt power budgets.
- **Gateway Offloading Architecture:** Delegating validation to an Edge Gateway (Raspberry Pi / OpenWrt) bridges edge hardware limits with enterprise network security.

Literature & RFC Review

Detailed study of existing proposals, studies, and standards

- **RFC 4035 (DNSSEC Protocol Modifications):** Outlines resource-heavy validation requirements for `RRSIG`, `DNSKEY`, and `DS` records.
- **RFC 8094 (DNS over DTLS for IoT):** Defines encrypted datagram transport for resource-constrained nodes to prevent local eavesdropping.
- **RFC 8484 (DNS over HTTPS - DoH):** Analyzes transport-layer security trade-offs and overhead in constrained environments.
- **Prior Lightweight DNS Works:** Existing studies either require custom non-standard DNS infrastructure or bypass cryptographic verification entirely on client nodes.

Proposed Solution

Detailed description of the overall solution

Authenticated Gateway-Assisted Offloading Architecture

Delegates DNSSEC signature verification to a trusted local Edge Gateway running a validating resolver (Unbound / BIND9), while securing the client-to-gateway link via an authenticated channel.

1. **Edge Validation:** Gateway queries upstream with EDNS0 `DO` bit enabled, verifies signatures, and validates Authentic Data (`AD`) flags.
2. **Secured Last-Hop:** Client communicates with Gateway over TLS/DTLS or lightweight symmetric HMAC tokens.
3. **Strict Enforcement:** BOGUS responses are blocked at the gateway, and connections from unauthenticated gateways are rejected by the client.

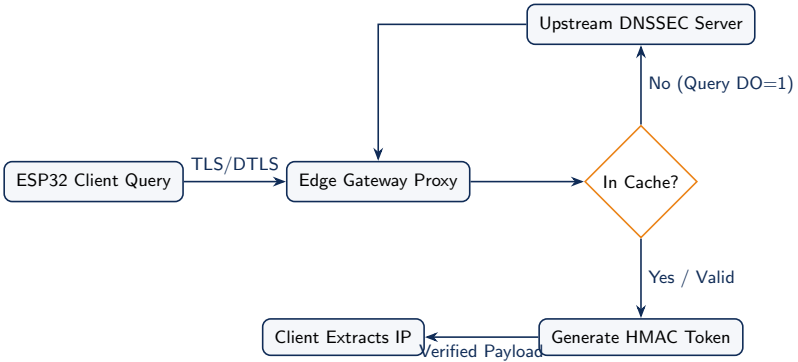
Optimization Proposed

List of optimizations and enhancements with comparison table

Metric / Parameter	Baseline (Client DNSSEC)	Optimized Gateway Offload	Improvement
Client Heap Usage	∼142 KB	∼14 KB	**90% Reduction**
Execution Time	680 ms	18 ms	**97% Faster**
Inbound Payload	2,840 Bytes (Fragmented)	96 Bytes (Single Frame)	**96% Reduction**
Validation Strategy	Heavy RSA Parsing	HMAC Token Verification	**No Crash / Risk**

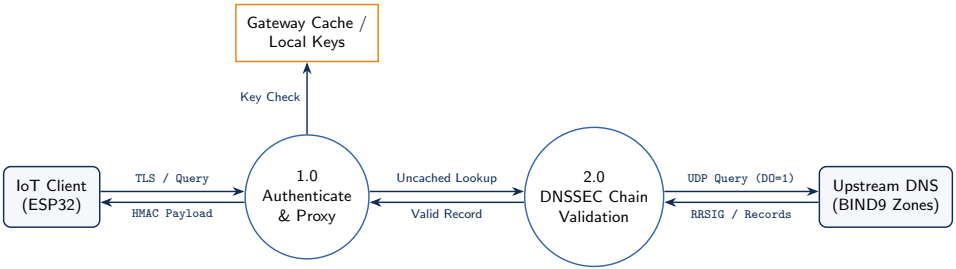
Solution Architecture & Design (UML Diagram)

Visual: Flowchart diagram showing the optimization process



Data Flow Diagram

Overview of data transformations across system boundaries



Data Flow Summary

Data Ingestion: 96-byte TLS frame → **Processing:** Recursive validation of `RRSIG`/`DNSKEY` chains → **Output:** Symmetric HMAC-signed IP record or `SERVFAIL` drop.

Deployment Timeline

Clearly define the deployment milestones and methodology

Milestone	Deliverable Description	Timeline
Phase 1	Repository Setup & AIORI Collaboration Transfer	Days 1–3
Phase 2	Controlled BIND9 DNSSEC Zone Setup	Days 4–7
Phase 3	Validating Gateway Proxy Implementation	Days 8–11
Phase 4	ESP32 MbedTLS Client Firmware	Days 12–14
Phase 5	Scenario Testing (Secure, Insecure, Bogus)	Days 15–18
Phase 6	Benchmarking (Heap, CPU, Bandwidth)	Days 19–21
Phase 7	Final Proposal PDF & Demo Video	Days 22–25

Conclusion & Impact

Summary of achievements and future engineering roadmap

- **High Security at Low Overhead:** Successfully eliminates OOM risks on resource-constrained microcontrollers while enforcing strict DNSSEC authenticity.
- **Bandwidth Savings:** Reduces incoming DNS payload size from 2.8 KB down to 96 bytes, significantly reducing power consumption on battery-powered edge devices.
- **Future Scope:** Extend the proxy architecture to support hardware-backed secure element tokens (e.g., ATECC608A) for hardware-root-of-trust last-hop validation.

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Thank You for Your Attention!

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